

An Analysis Across Huntington's Disease Stages: Peripheral Blood HLA-DQA1 Expression

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Introduction

Huntington's disease (HD) is an inherited neurodegenerative disorder caused by a CAG-repeat expansion in the HTT gene, resulting in progressive cognitive, psychiatric, and motor symptoms. Peripheral blood gene expression offers a less invasive approach to investigating disease-associated molecular changes, although previous research has found weak correlations with HD progression [1].

This project used Python to compare gene expression across five HD-stage groups. Among 54,675 probe sets, probe 213831_at (HLA-DQA1) showed the largest range between group means. The research question was: How does peripheral blood HLA-DQA1 expression vary across HD stages?

Data sample

The GSE135589 dataset was obtained from NCBI GEO [2]. It contains 178 peripheral whole-blood samples and 54,675 probe sets measured using Affymetrix microarrays [3]. Samples were classified as Control (n = 44), preHD A (n = 32), preHD B (n = 33), zHD stage 1 (n = 34), and zHD stage 2 (n = 35). Expression values were log2-transformed and RMA-normalized.

Python [4] was used for the analysis, with NumPy [5] for numerical calculations, pandas [6] for data processing, and Matplotlib [7] for visualization. For every probe, the mean expression of each group was calculated, and the probe with the largest range between group means was selected. Each group was divided into five subsamples (consecutive in order, not randomized) and the mean and population standard deviation (SD) of these five subsample means were calculated.

Results and discussion

The expression distributions (Figure 1) showed relatively stable medians across disease stages (12.09–12.64), a range of 0.55 log2 units. In contrast, group means varied by 2.59 units. The relatively stable medians and larger differences between group means suggest that lower expression values may have influenced the averages. This is also visible in Figure 1: in preHD A, preHD B, and zHD stage 2, the lower quartile lies at approximately 4.4, showing that a substantial proportion of samples had relatively low expression values.

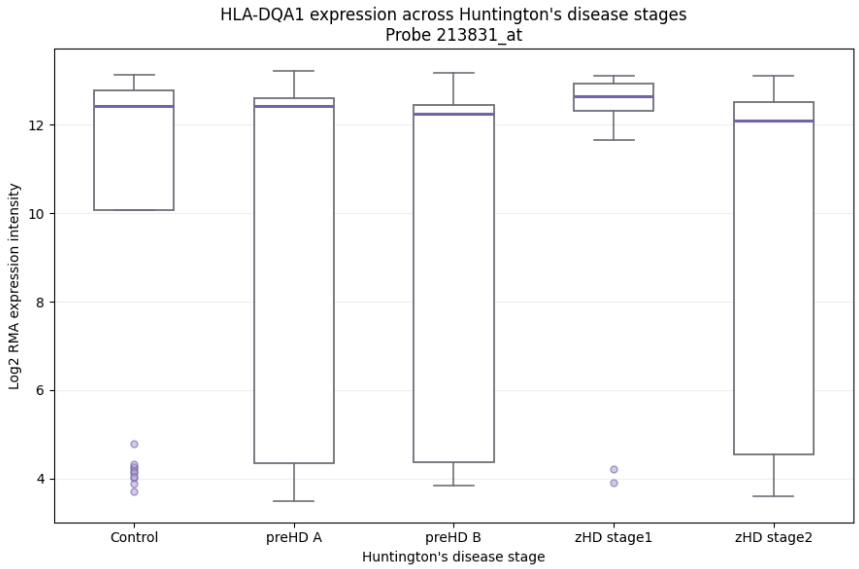


Figure 1: Distribution of *HLA-DQA1* expression across HD stages. Boxes represent interquartile ranges, lines indicate medians, and points indicate outliers.

Mean expression (Figure 2) was 10.41 in Control, 10.07 in preHD A, 10.02 in preHD B, 12.15 in zHD stage 1, and 9.56 in zHD stage 2. The largest difference occurred between symptomatic stages 1 and 2 (2.59 units), whereas differences between Control and preHD groups were smaller (0.34–0.39 units). Subsample variability was lowest in zHD stage 1 (SD = 1.00) and highest in zHD stage 2 (SD = 1.97). Thus, expression did not follow a consistent progression-related trend, and the error bars cannot establish statistical significance. Because each group was divided into only five subsamples, the error bars describe variation between these subsample averages rather than uncertainty in the true population mean. In addition, the analysis did not account for possible repeated measurements from the same participants, meaning that the samples cannot necessarily be treated as independent observations.

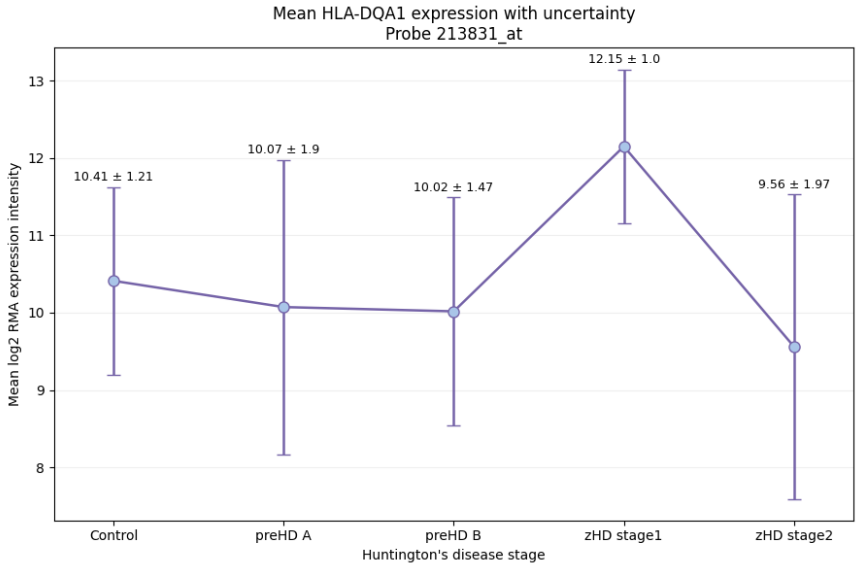


Figure 2: Mean *HLA-DQA1* expression across HD stages. Error bars represent the standard deviation of five subsample means, not confidence intervals. The line connects group means for readability and does not imply a trend.

Although *HLA-DQA1* participates in immune antigen presentation, its biological relevance to HD remains uncertain. Selecting the largest difference among 54,675 probes introduces selection bias, while repeated measurements were not explicitly modeled. The smaller median range relative to the mean range suggests that low-expression observations influenced group comparisons. The apparent separation between low and high expression values may indicate different expression patterns between samples rather than a gradual change with disease progression. However, this cannot be confirmed from the box plots alone.

In summary, *HLA-DQA1* expression did not show a consistent increase or decrease across HD stages. While median expression remained relatively stable, mean expression was highest in zHD stage 1 and lowest in zHD stage 2, with differences appearing to be influenced by low-expression observations. Although *HLA-DQA1* showed the largest range between group means among 54,675 probes, this exploratory analysis does not establish it as a biomarker of HD progression. Independent validation and formal statistical testing would be required.

References

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